

Construction of the Cumulative Wetland Effectiveness Variable W

Methods and Literature Review — from mechanism to a computable, calibrated metric

MMSD Wetland–Flood Effectiveness Study (wetland_v2)

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Abstract

This note documents, end to end, how we construct the cumulative “wetland effectiveness” predictor W_j used in the event-based flood-response models, and—equally—*why* each modeling choice is made, with every choice anchored to the hydrologic literature. Starting from the physical mechanisms by which wetlands modify floods (storage, attenuation, delay, infiltration, connectivity), we adopt the advisor’s cumulative form $W_j = \sum_i A_i S_i K(T_{ij})$, drop the separate connectivity term as redundant with the travel-time kernel, and build each remaining factor on a published method: a velocity/celerity time-of-concentration field for T_{ij} , a Volume–Area storage law and a Cowardin/HGM type weight for S_i , and an exponential travel-time kernel for K . We then calibrate the travel-time engine against observed event lags. The resulting variable is physically defensible and, reassuringly, reproduces the same peak-shaving signal as the simpler original metric.

1 Motivation: turning many wetlands into one number

The scientific question is whether the configuration of wetlands upstream of a gauge is associated with the flood response measured at that gauge. This requires collapsing thousands of heterogeneous wetland polygons into a single, physically meaningful predictor per gauge j . The guiding principle (Discussion note & workflow, 2026) is to *avoid inventing variables*: every quantity is borrowed from an authoritative published method. Following the discussion note, we adopt the cumulative “wetland effectiveness” form

$$W_j = \sum_{i \in \text{basin}(j)} A_i C_i S_i K(T_{ij}), \quad (1)$$

where A_i is the area of wetland i , C_i a connectivity/near-stream indicator, S_i a storage/soil/type modifier, T_{ij} the travel time from wetland i to gauge j , and $K(\cdot)$ a travel-time decay function. Equation (1) is a weighted sum: each wetland contributes its area, up-weighted or down-weighted by how much it can plausibly attenuate the flood and by how strongly it is connected to the gauge.

2 Literature review: mechanisms $\rightarrow$ mathematical representation

Wetlands do not affect floods through a single process, and the literature is explicit that the *type* and *location* of a wetland govern which process dominates (Acreman & Holden, 2013; Bullock &

Table 1: Wetland flood mechanisms and their representation in W_j .

Mechanism	Hydrologic effect	Representation (and literature)
Storage	Lower runoff volume; reduced peak	storage capacity V_i via Volume–Area (Gleason et al., 2007; Lane & D’Amico, 2010; Wu & Lane, 2016)
Attenuation	Lower peak, flatter hydrograph	travel–time weighting $K(T_{ij})$ (Acreman & Holden, 2013)
Delay	Longer time to peak	travel–time weighting / lag (McMillan, 2020)
Infiltration	Lower runoff coefficient	soil / wetland–type modifier in S_i
Connectivity	Stronger effect for near/connected wetlands	<i>is</i> the travel–time weighting (Lane et al., 2018; Ameli & Creed, 2019)
Peak–shaving	Disproportionate reduction in large storms	interaction $W \cdot P$ (model, not W)

Acreman, 2003). Table 1 maps each mechanism to its hydrologic effect and to the mathematical object we use to represent it; the right–hand column is the design logic for Eq. (1).

Three results from this literature drive our specific choices. First, Acreman & Holden (2013) review shows attenuation and delay dominate the flood signal (wetlands change *timing and shape* more than total volume), justifying a travel–time–weighted metric rather than a plain area. Second, Ameli & Creed (2019) demonstrate with a physically based model that wetlands within ~ 100 m of the channel attenuate peaks far more than wetlands >4 km away—i.e. the effect *decays with hydrologic distance to the network*, which is exactly a travel–time kernel. Third, Fossey & Rousseau (2016) and Evenson et al. (2015) show the location effect interacts with type (riparian wetlands matter most on the main stem; isolated wetlands act as headwater “gatekeepers” (Golden et al., 2016)), motivating a wetland–type modifier S_i .

3 Design decision: dropping the separate connectivity term C_i

Equation (1) contains both a connectivity indicator C_i (e.g. a <100 m near–stream flag) and a travel–time kernel $K(T_{ij})$. These encode the *same* mechanism twice: a near–stream wetland necessarily has a small T_{ij} and therefore a large $K(T_{ij})$. The authoritative connectivity framework itself defines connectivity operationally as travel time to the network (Lane et al., 2018), and Ameli & Creed (2019) express the “ <100 m matters most” finding as a continuous decay with distance, not a binary. Retaining both double–counts connectivity and inflates the weight of near–channel wetlands. We therefore drop C_i and let $K(T_{ij})$ carry all of the connectivity information:

$$W_j = \sum_i A_i S_i K(T_{ij}) \quad (2)$$

This is more parsimonious and removes a redundant, hard–to–calibrate parameter.

4 Constructing each factor

4.1 Wetland area A_i

Wetland polygons are the USFWS National Wetlands Inventory (NWI), carrying the Cowardin classification code. We query them by catchment, deduplicate, and clip; A_i is the clipped polygon

area in an equal-area projection. As a validation, the NWI wetland area of the main Milwaukee basin (324 km²) matches an independent NLCD estimate (325 km²).

4.2 Travel time T_{ij} : a velocity/celerity time-of-concentration field

Why travel time, not distance. Straight-line distance misranks wetlands because runoff follows the channel network; a small wetland on the stream can matter more than a large one off to the side. Travel time along the flow path is the physically correct connectivity measure (Lane et al., 2018). We compute it with the NRCS *velocity method* of time of concentration (NRCS, 1986; McCuen, 2009): route the 10 m DEM by D8 flow directions, assign each cell a flow speed, and sum L/speed along the path from every cell to the gauge (`pysheds distance_to_outlet`).

DEM conditioning. On the low-relief glacial terrain of southeastern Wisconsin, plain depression filling fragments the drainage. We use WhiteBox least-cost breaching (`BreachDepressionsLeastCost`) so the contributing area to each gauge matches its mapped basin, then `resolve_flats` to define D8 directions on flats.

Speed 1 — celerity, not water velocity. The original model used the Manning water velocity $v = \frac{1}{n} R^{2/3} S^{1/2}$. But a flood *peak* propagates at the kinematic wave *celerity*, $c = \beta v$, with $\beta = 5/3$ for a wide Manning channel (Lighthill & Whitham, 1955; Beven, 2012). Because we weight wetlands by how their *signal* reaches the gauge, celerity is the correct speed; using the water velocity systematically overestimates the travel time. Adopting $c = \frac{5}{3}v$ in channel cells de-inflates the modeled travel times by $\approx 1.5\times$ (e.g. basin-median 479 $\rightarrow$ 297 h at the main gauge) while preserving the spatial pattern.

Speed 2 — wetland roughness. With a uniform Manning n , a wetland does not actually slow water in the simulation, so the metric would encode only wetland *position*. Yet emergent marsh has a much higher roughness ($n \approx 0.05\text{--}0.15$) than upland surfaces (Kadlec, 1990; Arcement & Schneider, 1989). We therefore assign a land-cover-dependent overland resistance, slowing flow through wetland cells. Now the flow-path travel time *integrates the roughness of the wetlands the water passes through*: T_{ij} reflects wetlands actively delaying the flood, turning W from a position index into a mechanism.

4.3 Kernel $K(T_{ij})$ and the decay scale τ

We use the exponential kernel $K(T) = e^{-T/\tau}$ (with the harmonic $1/(1 + T/\tau)$ as a robustness variant): it is smooth, bounded, and reduces to the plain wetland fraction when the weight is constant (Discussion note & workflow, 2026). The decay scale τ is the travel-time distance over which a wetland’s influence persists. Because absolute travel times on this flat terrain are still long (hundreds of hours), a fixed τ (e.g. 24 h) would send nearly every weight to zero; we set τ adaptively to each catchment’s median travel time, which keeps the weighting well-scaled and comparable across basins, and calibrate it against data (§5).

4.4 Storage/type modifier S_i

The advisor’s S_i bundles storage, soil, and wetland type. We construct it so that the product $A_i S_i$ is mechanistically interpretable:

$$S_i = M_{\text{type},i} \times d_i, \quad A_i S_i = M_{\text{type},i} V_i. \quad (3)$$

Storage depth d_i (Volume–Area). Rather than hand-pick coefficients, we use the published Volume–Area power law $V_i = c A_i^\beta$ (Gleason et al., 2007; Lane & D’Amico, 2010; Wu & Lane, 2016), so the depth is $d_i = V_i/A_i = c A_i^{\beta-1}$. We adopt a glaciated–terrain fit ($c=0.01725$, $\beta=1.30$, cf. prairie–pothole depressions; Wu & Lane, 2016) appropriate to Wisconsin’s glacial landscape, replacing an earlier placeholder. Then $A_i S_i = M_{\text{type}} V_i$ and

$$W_j = \sum_i M_{\text{type},i} V_i e^{-T_{ij}/\tau}, \quad (4)$$

i.e. the connectivity–delivered, type–weighted *wetland storage* reaching gauge j . This unifies the storage capacity into W : $S_{w,j} = \sum_i V_i$ is the same quantity without the kernel.

Type modifier M_{type} . Following Acreman & Holden (2013), floodplain and emergent wetlands attenuate peaks more than open–water or lacustrine systems. We map the Cowardin class to an attenuation weight (emergent 1.0; forested/scrub–shrub 0.85; open water/aquatic bed 0.5; riverine 0.6; lacustrine 0.3)—a transparent, tunable table grounded in the type ordering of Acreman & Holden (2013); Bullock & Acreman (2003). Soil (saturated conductivity) is retained as a basin–level control rather than a per–wetland factor, because point SSURGO values on small polygons are noisy and infiltration acts at the catchment scale.

5 Calibrating the travel–time engine

A model of travel time should be checked against observed response times. From the event pipeline we have, per gauge, the observed hydrograph lag (time to peak). The SCS relation gives basin lag $\approx 0.6 T_c$ (NRCS, 1986). Regressing observed lag on the modeled travel time across the nine gauges yields lag $\approx 0.19 T_c$ ($R^2 = 0.46$): the model explains about half the cross–gauge variance, but the median travel time over–represents slow overland/wetland cells relative to the channel–dominated response, and the *flashy urban creeks fit worst*—their 2–3 h response is driven by imperviousness and storm sewers, not natural travel time. We therefore report the calibration as a validation and retain the adaptive τ ; a fully calibrated, urbanization–corrected celerity is flagged for future work.

6 Result and robustness

The improved variable of Eq. (4)—physical (celerity) travel times, wetland roughness, the redundant C removed, and a type–weighted glaciated storage S_i —reorders the wetland ranking (the most wetland–dense rural basin now scores highest per unit area). Crucially, when substituted into the event models it *reproduces the same peak–shaving signal* as the original connectivity–weighted fraction: the standardized $W \cdot P$ coefficient on log peak discharge is -0.155 (wild–cluster–bootstrap $p = 0.029$) versus -0.157 ($p = 0.004$) for the original metric. That the finding is invariant to a substantial, literature–grounded rebuild of W is itself evidence that it is not an artifact of one arbitrary construction.

7 What the literature says we still cannot do

Two honest limits remain. (i) *Identification.* W_j is gauge–constant and, across only nine nested catchments, collinear with urbanization; the wetland *level* effect is therefore not identifiable regardless of how well W is built—only the event–level interaction $W \cdot P$ is. A causal wetland level effect

needs a regional sample of independent catchments spanning a wetland gradient decoupled from urbanization. (ii) *Type*×*location tension*. Ameli & Creed (2019) (near-channel wetlands attenuate peaks most) and Fossey & Rousseau (2016) (isolated wetlands most effective at headwaters) are not fully reconciled; a single kernel cannot capture both, and separating riparian from isolated *W* components is the natural next refinement.

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